Joshua **Goodwill**

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Education

University of Delaware — *Newark, DE*

Ph.D. Candidate in Physics — *Expected Spring 2027*

Duquesne University — *Pittsburgh, PA*

Bachelor of Science in Physics — *May 2021*

Mathematics Minor, Computer Science Minor

Research Experience

University of Delaware Space Physics Group — *Newark, DE*

Research Assistant — *May 2021 - Present*

- **Advisor:** Prof. William H. Matthaeus
- Investigated energy dissipation in collisionless plasma turbulence, emphasizing nonlinear evolution and kinetic-scale processes
- Analyzed pressure–strain interaction to quantify energy conversion pathways in turbulent plasmas
- Analyzed Parker Solar Probe data to characterize the Alfvénic transition, including velocity shear, magnetic deflections, and switchback formation
- Performed statistical studies of solar wind turbulence, connecting observational signatures from plasma turbulence and reconnection

Los Alamos National Laboratory: T2-Plasma Astrophysics Group — *Los Alamos, NM*

Graduate Research Fellow (CSES Fellowship) — *May 2022 - Sept 2024*

- **Advisor:** Dr. Fan Guo
- Conducted large-scale particle-in-cell (PIC) simulations using VPIC to study shear-driven turbulence and reconnection in collisionless plasmas
- Designed and analyzed simulations to investigate energy dissipation, intermittency, and reconnection-driven heating
- Quantified statistical properties of reconnection sites and dissipation structures using kinetic simulations to connect with turbulence

Thomas Jefferson National Laboratory: Experimental Hall B — *Norfolk, VA*

Science Undergraduate Laboratory Internships (SULI) — *Summer 2019 & 2020*

- **Advisor:** Dr. Nathan Baltzell
- Developed real-time access to hardware performance for the CLAS12 Ring Imaging Cherenkov Detector (RICH)

Publications

Averaging Effects on the Solar Wind Alfven Mach Number: Implications for Switchbacks and Alfven Transition — *Submitted; JGR: Space Physics*

J. Goodwill, S. Adhikari, F. Pecora, R. Pradata, J. Wang, S. P. Gautam, M. E. Cuesta, O. Romeo, D. Ruffolo, P. Thepthong, P. Pongkitiwanichakul, R. Chhiber, W. H. Matthaeus

Parker Solar Probe Analysis across the Alfvénic Transition: Velocity Shear, Magnetic Deflection and Switchback Formation — *MNRAS 2026*

J. Goodwill, S. Adhikari, D. Payne, R. Bandyopadhyay, S. Badman, F. Pecora, P. Pongkitiwanichakul, R. Pradata, O. Romeo, S. Roy, D. Ruffolo, M. Stevens, P. Thepthong, A. Usmanov, J. Wang, M. Goldstein, R. Chhiber, W. H. Matthaeus

Evolution of Magnetic Deflections at a Conversion Layer near the Alfvén Surface — *ApJL 2026*

D. Payne, M. Akhavan-Tafti, **J. Goodwill**, S. Badman, R. Bandyopadhyay, G. Zank, S. Adhikari, W. Matthaeus, C. Shi, M. Stevens, R. Livi, Y. Rivera, K. Paulson

Characterization of the Alfvén Transition in the Young Solar Wind using Parker Solar Probe Observations Approaching Solar Maximum — *ApJ 2026*

S. Adhikari, R. Bandyopadhyay, **J. Goodwill**, W. H. Matthaeus, D. Ruffolo, P. Thepthong, P. Pongkitiwanichakul, S. Roy, F. Pecora, R. Chhiber, R. Pradata, A. Usmanov, M. Stevens, S. Badman, O. Romeo, J. Wang, M. Goldstein

Nonlinear Evolution and Energy Dissipation in Shear-Driven Collisionless Plasma Turbulence — *PoP 2025*

J. Goodwill, S. Adhikari, X. Li, F. Pucci, Y. Yang, F. Guo, W. H. Matthaeus

Observed Fluctuation Enhancement and Departure from WKB Theory in Sub-Alfvénic Solar Wind *ApJL 2024*

D. Ruffolo, P. Thepthong, P. Pongkitiwanichakul, S. Roy, F. Pecora, R. Bandyopadhyay, R. Chhiber, A. Usmanov, M. Stevens, S. Badman, O. Romeo, J. Wang, **J. Goodwill**, M. Goldstein, W. H. Matthaeus

Isotropization and Evolution of Energy-containing Eddies in Solar Wind Turbulence: Parker Solar Probe, Helios 1, ACE, WIND, and Voyager 1 *ApJL 2022*

M. E. Cuesta, R. Chhiber, S. Roy, **J. Goodwill**, F. Pecora, J. Jarosik, W. H. Matthaeus, T. N. Parashar, R. Bandyopadhyay

Scale-Dependent Statistics of Coherent Magnetic Structures in Early IMAP Observations *Submitted*

S. P. Gautam, F. Pecora, S. Adhikari, R. Bandyopadhyay, J. Wang, **J. Goodwill**, W. Taranto, R. Pradata, W. H. Matthaeus, D. J. McComas, E. Christian, N. Schwadron, M. Gkioulidou, T. S. Horbury, S. Raptis, D. Turner

A Student Sounding Rocket Experiment to Probe Electron Density in the Lower Ionosphere *AJP 2027; Accepted*

J. Goodwill, T. Tomaszewski, J. Bieber, C. P. Brown, A. Southwick, S. Auerbach, S. Banda, A. Bruce, M. C. Olais, M. Duey, R. Khan, C. McCleery, J. Rosado, A. Swenson, M. Ward, J. Dagney, D. Sundkvist, J. W. Bonnell, W. H. Matthaeus, B. A. Maruca

Honors & Awards

Delaware Space Grant Consortium (DESGC) Graduate Fellowship *Sept 2024 - Spring 2027*

Los Alamos National Lab (LANL) CSES Fellowship (Advisor: Dr. Fan Guo) *May 2022 - Sept 2024*

Conferences & Talks

PSP Turbulence and Dissipation Working Group *April 2026*

Talk: Characterizing the Alfvénic Transition in the Inner Heliosphere

APS Mid-Atlantic Section *Dec 2022*

Talk: Proton and Electron Heating in Turbulent Magnetic Reconnection

Solar Wind 17 *June 2026*

Poster: Parker Solar Probe analysis across the Alfvénic transition: velocity shear, magnetic deflection, and switchback formation

AGU25 *Dec 2025*

Poster: Statistics of Energy Dissipation Related to Reconnection in Turbulent Plasma

AGU24 *Dec 2024*

Poster: Nonlinear Evolution and Energy Dissipation in Shear-Driven Collisionless Plasma Turbulence

SHINE *June 2024*

Poster: Nonlinear Evolution and Energy Dissipation in Shear Driven Turbulence of Collisionless Plasma

AGU23 *Dec 2023*

Poster: Statistical Analysis of Energy Dissipation Around Reconnection Sites in a Turbulent Environment

Solar Wind 16 *June 2023*

Poster: Statistical Analysis of Energy Dissipation Around Reconnection Sites in a Turbulent Environment

SHINE *June 2022*

Poster: Isotropization and Evolution of Energy-Containing Eddies in Solar Wind Turbulence: Parker Solar Probe, Helios 1, ACE, WIND, and Voyager 1

2018 RICH Meeting *June 2018*

Talk: RICH calibration data

Relevant Skills

Codes/Software LANL VPIC, P3D, Athena++
Programming Python, C++, Mathematica

Student Mentorship

Delaware Atmospheric Plasma Probe Experiment (DAPPEr) *Aug 2023 - Nov 2025*

Jarrod Bieber, Carson Brown, Ashton Southwick, Owen Blume

- Utilized fundamental plasma physics to guide scientific development and mentorship on low-earth orbit Langmuir Probe mission

University of Delaware Ionospheric Probe (UDIP-4) *Aug 2023 - June 2024*

Marisol Catalan Olais, Amanda Swenson

- Led the science team for mission design and developed semester/summer projects for undergraduate students.
- Mission designs went through NASA's design reviews to ensure flight readiness. UDIP-4 was launched at NASA Wallops Flight Facility.

References

Prof. William H. Matthaeus *Martin A. Pomerantz Chair of Physics & Astronomy*

Doctoral Research Advisor *University of Delaware*

- whm@udel.edu

Prof. Riddhi Bandyopadhyay *Assistant Professor*

Research Collaborator *University of Delaware*

- riddhib@udel.edu

Dr. Fan Guo *Staff Scientist*

Los Alamos National Lab Research Advisor *Los Alamos National Laboratory*

- guofan@lanl.gov